
IZMIR INSTITUTE OF TECHNOLOGY
CENG 534 • SPRING 2026

Building *GPT-2* from Scratch

*Backbone, optimizer, and downstream adaptation - a CS 224N
default final project.*

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DATE

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COURSE

CENG 534

RUNTIME

≈ 14 minutes



IYTE

EST. 1992

Implement the *backbone*, replicate the *optimizer*, and adapt to three downstream tasks.

PILLAR I

Backbone

Attention · pre-LN block · embeddings · weight-tied LM head.

PILLAR II

Optimizer

AdamW - bias correction + decoupled weight decay.

PILLAR III

Validation

Logits match HuggingFace GPT-2 (124M).

THREE DOWNSTREAM TASKS

Task 1 - Sentiment

SST-5 & CFIMDB.

Task 2 - Paraphrase

Quora question pairs.

Task 3 - Sonnet

Shakespearean sonnets · chrF + BLEU.

EXTENSION · PART 3 LoRA (PEFT) - low-rank adapters on Q/K/V attention

COMPLETE

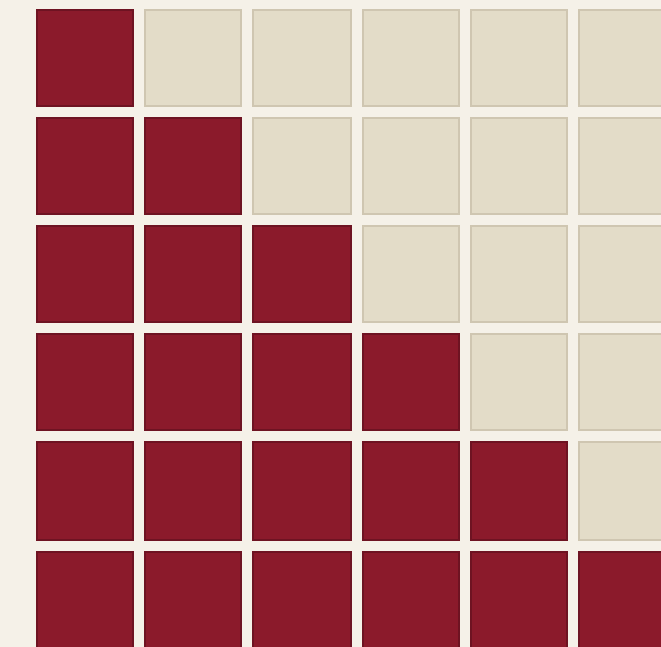
Causal self-attention: *scaled dot-product* with two additive masks.

FORMULA • MODULES/ATTENTION.PY

```
Attn( $Q$ ,  $K$ ,  $V$ ) =
    softmax(  $Q K^T / \sqrt{d_k}$ 
            +  $M_{\text{causal}}$ 
            +  $M_{\text{pad}}$  )  $V$ 
```

- **Causal mask** - future positions $\rightarrow -10^4$ before softmax.
- **Padding mask** - variable-length batches.
- *Matches HuggingFace GPT-2 logits.*

CAUSAL MASK • 6×6



POSITION I ATTENDS TO $J \leq I$

Pre-LayerNorm block, GELU feed-forward, weight-tied LM head.

BLOCK UPDATE · MODULES/GPT2_LAYER.PY

```
h' = h + Dropout( W0 · Attn( LN1(h) ) )

h'' = h' + Dropout( W2 · GELU( W1 · LN2(h') ) )
```

- **Pre-LN** - better gradient flow, no warm-up needed.
- **Weight tying** - logits = $h \cdot E_{\text{token}}^T$.
- *≈ 38 M parameters saved.*

LAYER · FORWARD PASS



AdamW: bias-corrected moments with *decoupled* weight decay.

OPTIMIZER.PY • 32 LINES

```

1)  $m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$ 
2)  $v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$ 

3)  $\alpha_t = \alpha \cdot \sqrt{(1 - \beta_2^t)} / (1 - \beta_1^t)$ 

4)  $\theta_t \leftarrow \theta_{t-1} - \alpha_t \cdot m_t / (\sqrt{v_t} + \epsilon)$ 
5)  $\theta_t \leftarrow \theta_t \cdot (1 - \alpha \lambda)$  ▶ decoupled

```

- **Efficient bias correction** - one step-size scalar α_t , not per-step normalisation.
- **Decoupled weight decay** - applied to parameters, never mixed into the gradient.
- *Matches reference, $atol = 1 \times 10^{-6}$.*

Sentiment classification: after tuning, *both regimes* match or exceed the baseline.

DEV ACCURACY · OURS VS. BASELINE

SETUP	OURS	BASELINE	Δ
SST · last-linear	0.455	0.462	-0.7 pts
SST · full-model	0.508	0.513	-0.5 pts
CFIMDB · last-linear	0.878	0.861	+1.7 pts ▲
CFIMDB · full-model	0.984	0.976	+0.8 pts ▲

last-linear lr = 1e-3 · full-model tuned (lr / epochs / batch)

RESOLVED BY TUNING

SST full-model gap
-9.9 pts → -0.5 pts

Tuning lr · epochs · batch and adding dropout closed the SST gap; CFIMDB full-model now exceeds baseline.

FULL-MODEL TUNED · FINAL

Paraphrase detection on Quora: our forward pass connects GPT-2 to the classification head.

FORWARD PASS · PARAPHRASE_DETECTION.PY

```
# encoder → last-token hidden → binary head
outputs = self.gpt(input_ids, attention_mask)
return self.paraphrase_detection_head(
    outputs['last_token'])
```

- **Starter:** prompt template + Linear(768 → 2) head.
- **Ours:** full-parameter fine-tune · AdamW lr = 1e-5 · CE on 0/1.

DEV ACCURACY

≈ 0.896

on 40,429 dev examples

Sonnet generation: forward, training, decoding and chrF all *integrated end-to-end*.

FORWARD PASS · SONNET_GENERATION.PY

```
# per-token logits via weight-tied LM head
outputs = self.gpt(input_ids, attention_mask)
hidden = outputs['last_hidden_state']
return self.gpt.hidden_state_to_token(hidden)
```

- **Train** - CE on 143 sonnets.
- **Decode** - top-p = 0.9 · T = 1.2.
- **Eval** - chrF + BLEU after every epoch.

HELD-OUT SONNET · PROMPT + COMPLETION

*Shall I compare thee to a summer's day?
Thou art more lovely and more temperate:
Rough winds do shake the darling buds of May,*

GENERATED

*And her lips stretching in reply
next to comfort me ...*

CHRF

40.95

BLEU

23.75

held-out dev · 12 sonnets

LoRA: freeze the backbone, inject a low-rank adapter into every attention projection.

ADAPTER · MODULES/ATTENTION.PY

$$h = W \cdot x + (a/r) \cdot B(A \cdot x)$$

A: hidden $\rightarrow$ r B: r $\rightarrow$ all_head_size
no bias · $B = 0$ at init ▶ adapter starts as a no-op

- **Where** - Q, K, V across all 12 layers $\rightarrow$ **36 adapter pairs**.
- **Default** - $r = 8 \cdot \alpha = 16$ · scaling $\alpha/r = 2$.
- **Default-OFF** - `--lora_r 0` disables adapters; baselines stay intact.

TRAINABLE PARAMETERS

0.44M

of 124 M · 0.36 %

FROZEN

All pretrained GPT-2 weights - only `lora_*` require grad.

Initialization ablation: *default*, *weight-dist*, and *SVD* - measured by chrF + BLEU.

THREE INIT METHODS

- **default** - $A \sim \mathcal{N}(0, 0.02)$, $B = 0$ · *standard LoRA*
- **weight_dist** - $A, B \sim$ pretrained W distribution $\times$ scale
- **svd** - from the truncated SVD (Singular Value Decomposition) of W

We log adapter weight statistics every epoch to compare how each init shapes the adapters.

INIT METHOD × HELD-OUT DEV

SETUP	TRAIN %	CHRF	BLEU
Full fine-tune · ref	100 %	40.95	23.75
Last-layer	5.7 %	36.11	21.58
LoRA · default	0.36 %	39.29	23.07
LoRA · weight_dist	0.36 %	40.09	23.54
LoRA · svd	0.36 %	39.13	23.10

FINDINGS

LoRA reaches ~98 % of full fine-tune's chrF with ~0.4 % of its parameters and surpasses the last-layer probe - **weight_dist** init is the best PEFT variant.

The same LoRA ablation on *paraphrase*: three inits recover most of full fine-tuning at 0.36 % of the parameters.

PROTOCOL

- **Setups** - full fine-tune vs LoRA default / weight_dist / svd
- **Same config** - $r = 8 \cdot \alpha = 16 \cdot 0.44$ M trainable (0.36 %)
- **Metric** - dev accuracy on 40,429 Quora pairs

Contrast - the inits are indistinguishable here; on sonnet generation weight_dist led.

INIT METHOD × QUORA DEV ACC

SETUP	TRAIN %	DEV ACC	VS. FULL
Full fine-tune · ref	100 %	0.896	—
LoRA · default	0.36 %	0.845	-5.1 pt
LoRA · weight_dist	0.36 %	0.844	-5.2 pt
LoRA · svd	0.36 %	0.844	-5.2 pt

FINDINGS

All three inits land within **0.1 pt** of each other (84.4–84.5 %), recovering most of full fine-tuning at 0.36 % of the parameters. The spread is tight enough that **init choice is nearly irrelevant** on this task.

What we *built* and what we *validated*.

BUILT FROM SCRATCH

- ✓ Causal self-attention
- ✓ Pre-LN transformer block
- ✓ Token + position embeddings
- ✓ Weight-tied LM head
- ✓ AdamW with bias correction
- ✓ LoRA adapters on Q/K/V · Part 3

VALIDATED

- ✓ HuggingFace GPT-2 logits
- ✓ Reference optimizer step
- ✓ Sentiment full-model $\approx$ / $>$ baseline
- ✓ Paraphrase dev acc $\approx$ 0.896
- ✓ Sonnet chrF 40.95 · BLEU 23.75

Thank you.

QUESTIONS WELCOME